

# LogiSim Simulation Tool Guide

لاجی سیم ایوولوشن (LogiSim Evolution) لاجک سرکٹ بنانے اور چلا کر دیکھنے کا سافٹ ویئر ہے۔

CS XI

Chapter 1: Computer Systems

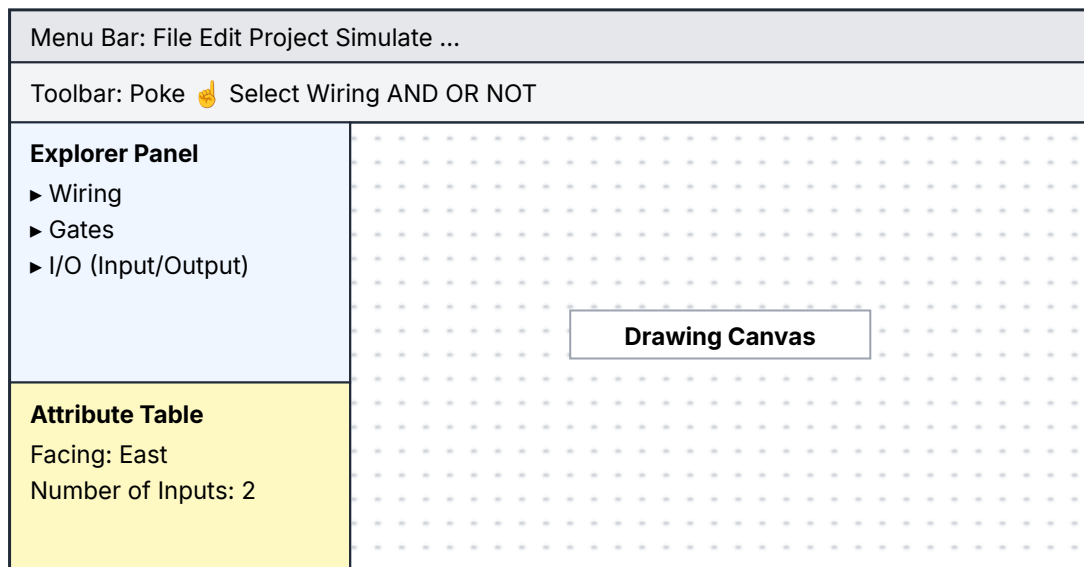
Lecture 08 of 12

## LogiSim Simulation Tool Guide

### LEARN IT

**LogiSim Evolution** is a free graphical tool for **designing and simulating logic circuits**. "Simulating" means the computer pretends to be the real circuit: you can switch inputs on and off and watch the output change. It lets you see the truth tables and Boolean expressions you learn in theory.

### 1. The interface areas



The main areas of the LogiSim Evolution window.

| Area                  | What it does                                                                      |
|-----------------------|-----------------------------------------------------------------------------------|
| Menu Bar (top)        | File, Edit, Project, Simulate menus — open/save files and control the simulation. |
| Toolbar               | Quick access to standard tools: Wiring, Poke, and AND, OR, NOT gates.             |
| Explorer Panel (left) | Contains the component libraries: Gates, I/O, Wiring.                             |

|                               |                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------|
| Attribute Table (bottom left) | Adjust the properties of the selected gate, e.g. Number of Inputs or Facing direction. |
| Drawing Canvas (right)        | The dotted grid where you place and wire the gates.                                    |

## Wiring colours during simulation

| Colour       | Meaning                   |
|--------------|---------------------------|
| Bright green | Logic 1 (HIGH)            |
| Dark green   | Logic 0 (LOW)             |
| Blue         | Unknown / unconnected pin |
| Red          | Error / short circuit     |

### REMEMBER IT

**Colours like a traffic light:** Green means "working" (bright = 1, dark = 0). Blue means "the wire is lonely — not connected". Red means "stop, something is wrong".

### COMMON MISTAKE

If your output wire is blue, the circuit is not wrong in logic — a wire is simply not touching a pin. Zoom in and make sure every wire end sits exactly on a connection point.

### اردو میں سمجھیں

لاجی سیم ایوولیوشن (LogiSim Evolution) لاجک سرکٹ بنانے اور چلا کر دیکھنے کا سافٹ ویئر ہے۔ اس میں مینو بار، ٹول بار، ایکسپلورر پینل، ایٹری بیوٹ ٹیبل اور ڈرائنگ کینوس ہوتے ہیں۔ چمکدار سبز تار 1، گہری سبز 0، نیلی تار غیر منسلک اور سرخ تار غلطی کو ظاہر کرتی ہے۔

### CHECK YOURSELF

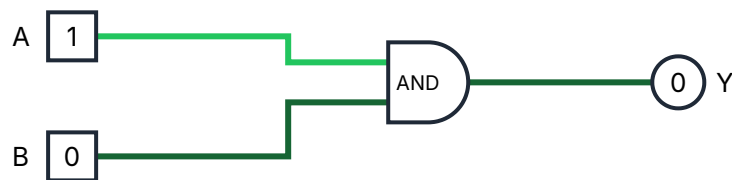
1. Where do you change the number of inputs of a gate?
2. What does a red wire mean?

Answers: 1. In the Attribute Table (bottom left). 2. An error or short circuit.

### WORKED EXAMPLE

#### Step by step: building $Y = A \cdot B$

1. Select the **AND gate** from the Toolbar (or from the Gates library) and place it on the canvas.
2. Select the **Input Pin** (square icon with a 0) and place two pins to the left of the gate. Label them A and B.
3. Select the **Output Pin** (circle icon with a 0) and place it to the right. Label it Y.
4. Use the **Wiring Tool** to click and drag wires from the input pins to the AND gate's input ports. Wires must be horizontal or vertical.
5. Wire the gate's output to the output pin Y.
6. Select the **Poke Tool** (pointing finger) and click inputs A and B to toggle them between 0 and 1. Test the circuit!



Bright green = 1, dark green = 0

Testing with the Poke Tool: A = 1 (bright green wire), B = 0 (dark green), so  $Y = 1 \cdot 0 = 0$ .

### WORKED EXAMPLE

**Testing the circuit = making its truth table.** Poke the inputs through all four combinations and write what Y shows:

| Poke A | Poke B | Y shows | Output wire colour |
|--------|--------|---------|--------------------|
| 0      | 0      | 0       | Dark green         |
| 0      | 1      | 0       | Dark green         |
| 1      | 0      | 0       | Dark green         |
| 1      | 1      | 1       | Bright green       |

This matches the AND truth table, so the circuit is correct. Try the same with  $Y = A \cdot B + C$  from 1.1.11 (add a second gate and a third input pin).

### REMEMBER IT

Input pin = **square** (you push it). Output pin = **circle** (it shows a light). Poke = finger — you "poke" inputs to change them.

### COMMON MISTAKE

Students often try to poke inputs while the Wiring or Select tool is active — then clicking moves the pin instead of changing it. Switch to the Poke Tool first.

### اردو میں سمجھیں

لاجی سیم میں  $Y = A \cdot B$  بنانے کے لیے AND گیٹ رکھیں، بائیں جانب دو ان پٹ پن (چوکور) اور دائیں جانب ایک آؤٹ پٹ پن (گول) رکھیں، پھر وائرنگ ٹول سے تاریں جوڑیں۔ آخر میں پوک ٹول (انگلی) سے ان پٹ کو 0 اور 1 کر کے سرکٹ کو جانچیں۔

### CHECK YOURSELF

1. Which icon is the input pin: square or circle?
2. Which tool do you use to change an input between 0 and 1?

Answers: 1. Square (output pin is the circle). 2. The Poke Tool (pointing finger).

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** LogiSim Evolution is used to:

- a. Draw and simulate logic circuits
- b. Edit videos
- c. Send email
- d. Compile Java only

**MCQ 2.** The canvas is where you:

- a. Place and wire gates
- b. Install Python
- c. Write SQL
- d. Draw pie charts

**MCQ 3.** Poke tool is for:

- a. Toggling inputs while simulating
- b. Deleting the OS
- c. Printing Urdu
- d. Sorting

**MCQ 4.** Attribute table sets:

- a. Inputs, facing, labels
- b. Big O
- c. IP addresses
- d. Passwords

**MCQ 5.** To build  $Y=A \cdot B$  you need:

- a. AND gate plus two inputs and an output
- b. Only XOR
- c. A stack
- d. Pandas

**MCQ 6.** Simulation lets you check:

- a. The truth table live
- b. HTTP status
- c. File modes
- d. HCI wireframes only

## Extra short questions

1. Name four areas of the LogiSim window.
2. Steps to build  $Y=A \cdot B$ .
3. Where do you change the number of inputs on a gate?
4. Why simulate at all if the truth table is already on paper?

## Extra long question

1. Describe, with a canvas sketch, how you would build and test  $Y = A \cdot B + C$  in LogiSim.

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

### Short

1. Menu bar, toolbar, explorer (libraries), attribute table, drawing canvas.
2. Place AND; add two input pins; add output pin; wire; label; simulate with Poke.

3. Attribute table (bottom left).

4. Catches wiring mistakes and shows students the table “come alive”.

**Long**

1. AND for A,B; OR with C; three inputs, one output; wiring; poke all 8 combinations; tick the truth table. Mention facing and labels.

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Next lecture: 1.2.1 – 1.2.3 Software Development Life Cycle (SDLC)